Title: *Predicting Student Performance Using Tree-Based Models*

1. Introduction

This project aims to predict whether a student will pass or fail based on various academic and personal attributes using machine learning models like Decision Tree and Random Forest. Such predictions can help educators provide early support to at-risk students.

---

2. Dataset Overview

- **Source**: Kaggle – Student Performance Dataset
- **Total Records**: 395
- **Features**:
  - Study time
  - Parental education
  - Absences
  - Past failures
  - Demographics
- **Target**: `G3` converted to `Pass` (1) or `Fail` (0)

---

3. Preprocessing

- Converted `G3` final grade into binary: Pass (≥10), Fail (<10)
- Encoded all categorical variables using LabelEncoder
- Split dataset: 80% training, 20% testing

---

4. Models Used

Model	Description
Decision Tree	Tree-based classifier (basic)
Random Forest	Ensemble of decision trees (advanced)

---

5. Evaluation Metrics

Decision Tree:

- Accuracy: 88.6%
- Precision: 92.1%
- Recall: 90.3%
- F1 Score: 91.3%

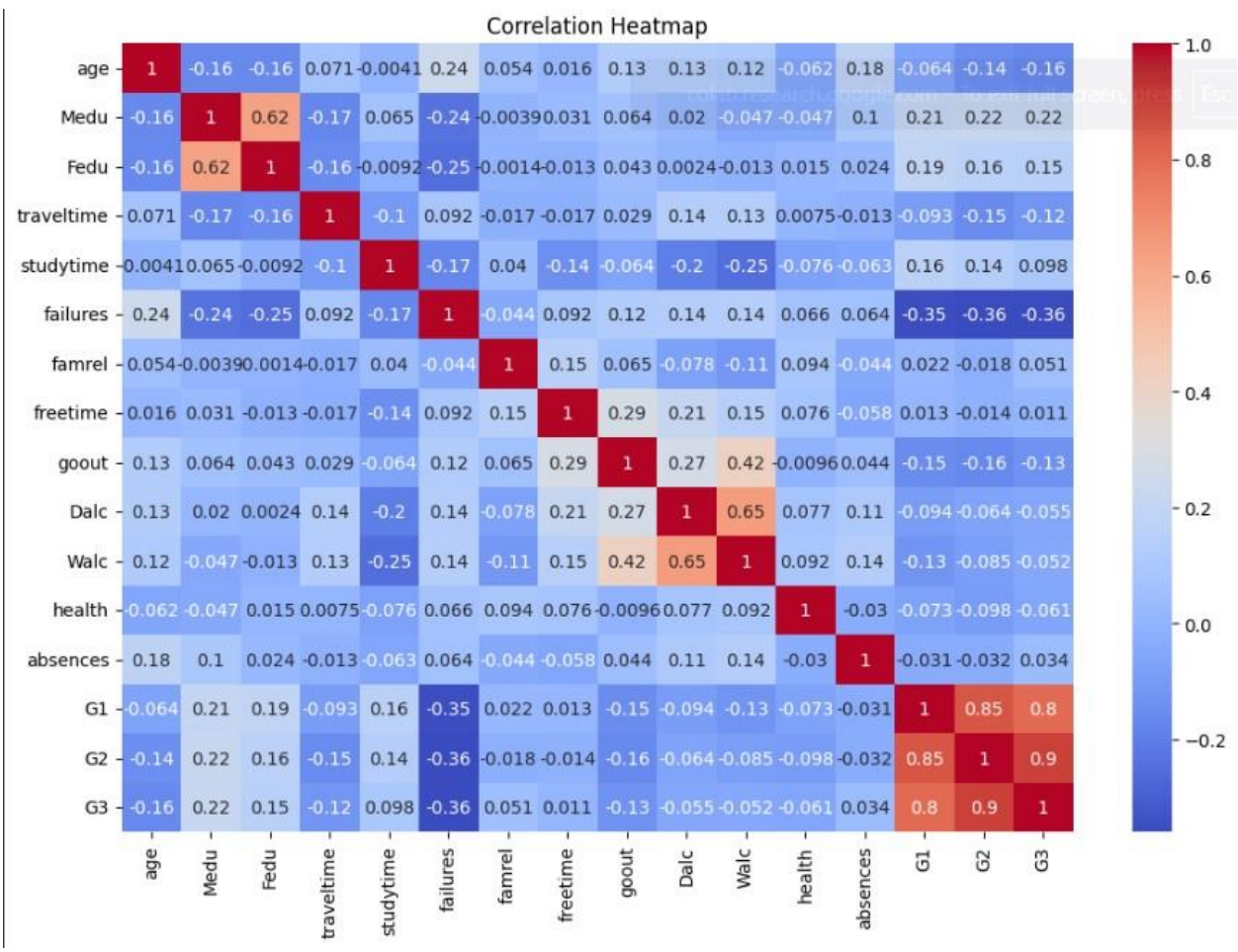
Random Forest:

- Accuracy: 91.1%
- Precision: 95.9%
- Recall: 90.3%
- F1 Score: 93.1%

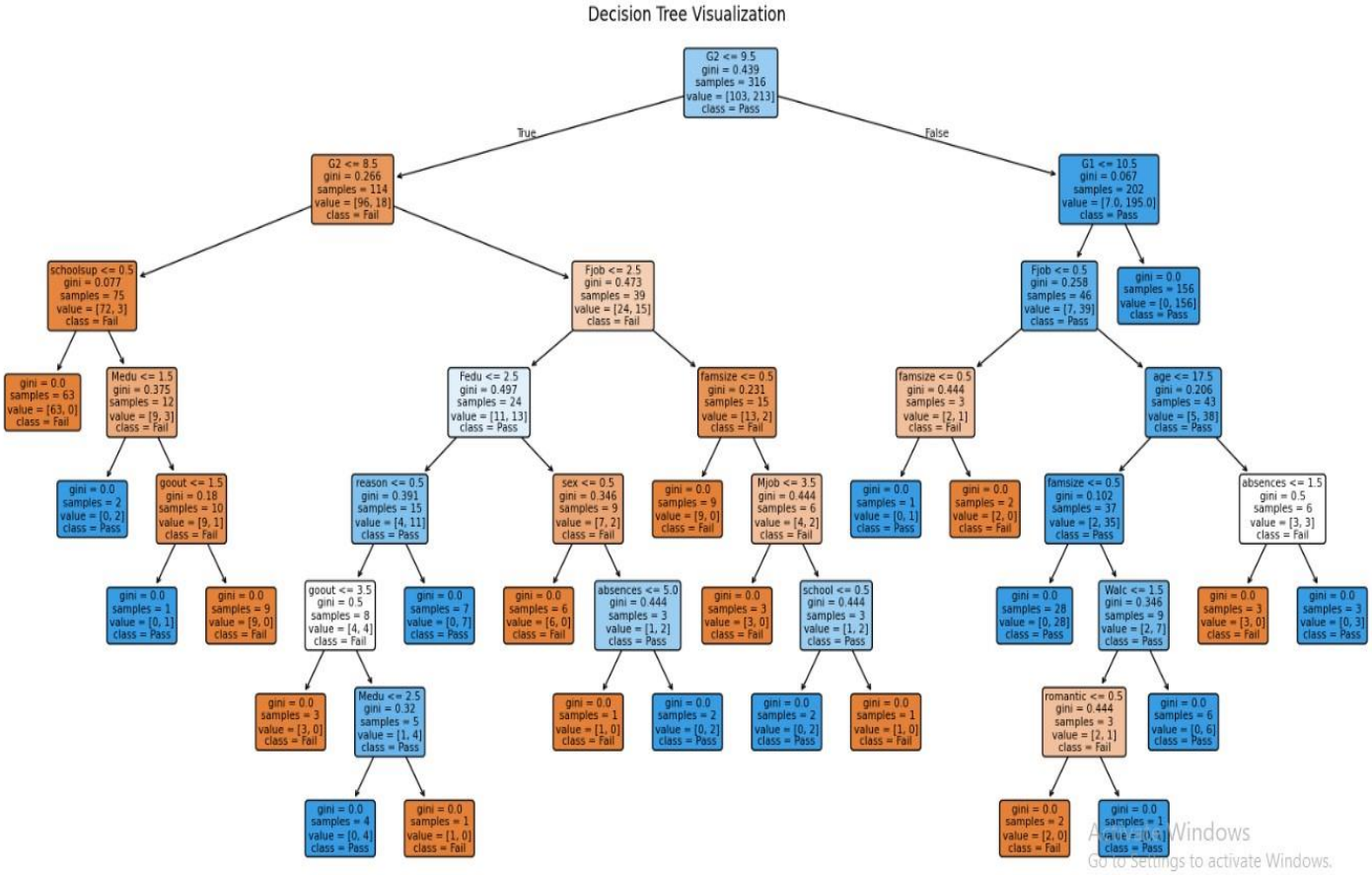
---

6. Visualizations

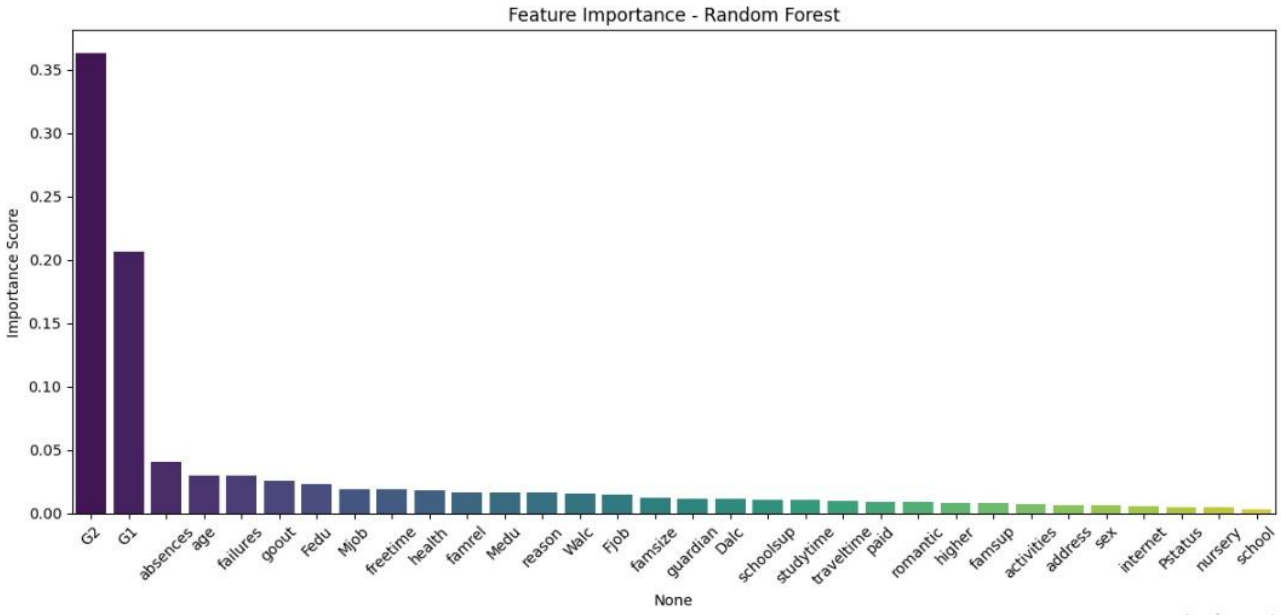
1.Correlation Heatmap :



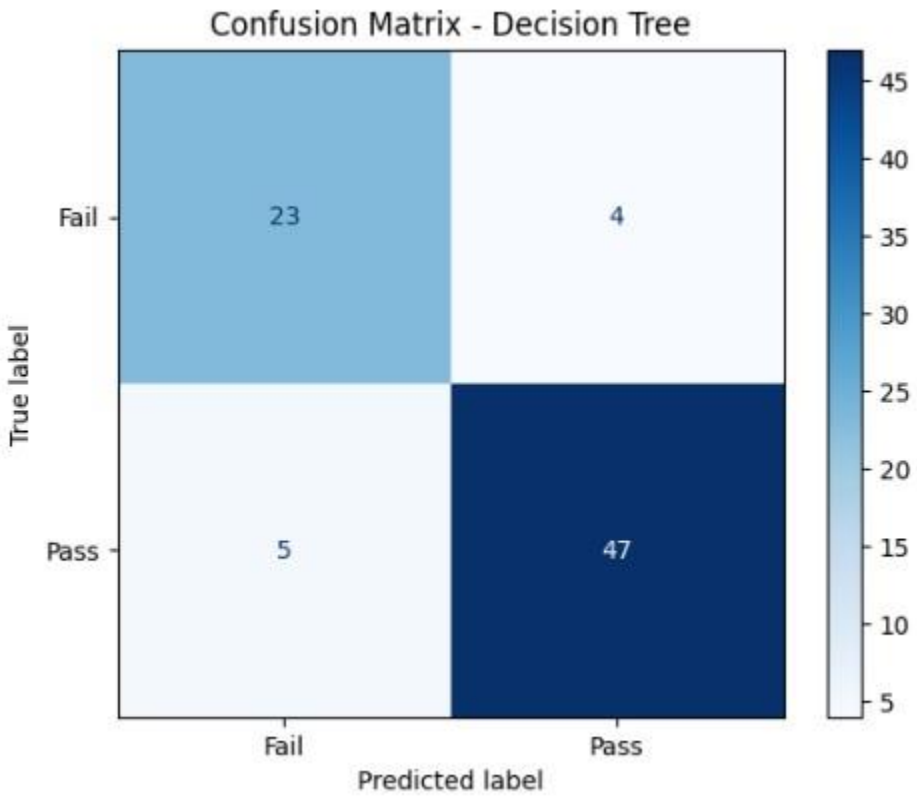
2.Decision Tree Plot :

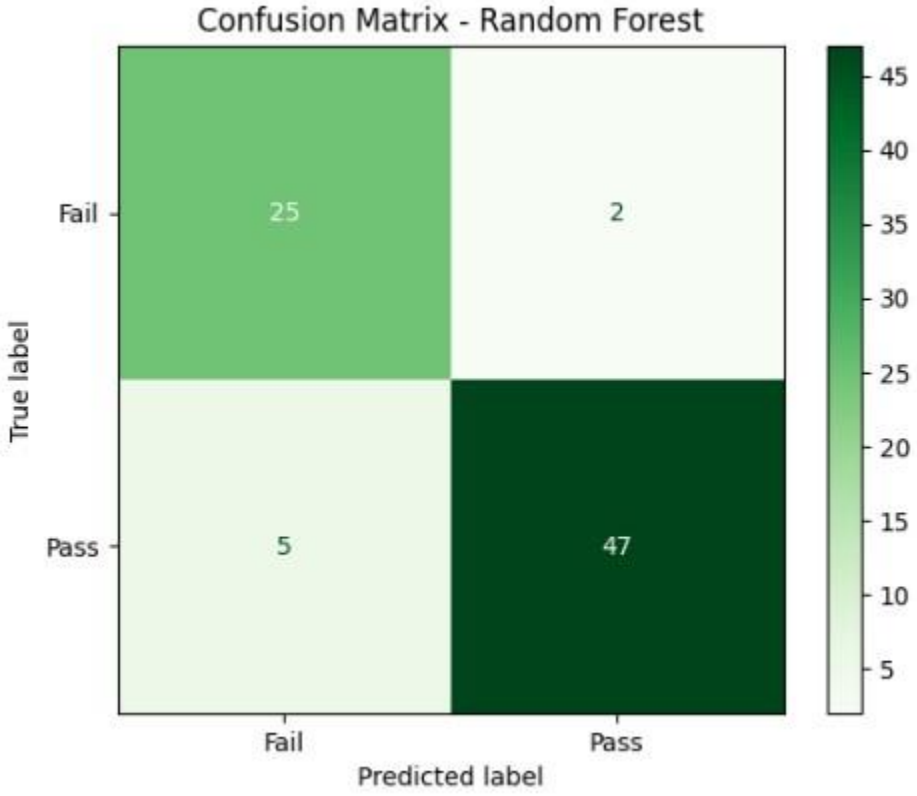


3.Random Forest Feature Importance ;



4.Confusion Matrix :





7. Insights

- **Random Forest** was the best-performing model.
- Important factors: Study time, past failures, absences
- Higher absences and low study time increase fail chances

---

8. Conclusion

Machine learning models like Random Forest can effectively predict student outcomes. This can help institutions identify students who need extra academic support.

---

9. Author

shaikh umair ahemad

Institute Name (mgm university)

domain(AIML)